

BUSINESS, LIQUIDITY, AND INFORMATION CYCLES*

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Stock markets play a dual role: they provide information about firms' fundamentals, which improves resource allocations, and they provide liquidity. We propose a setting in which these two roles interact: if stocks are used more intensively for liquidity, then prices reveal less information about fundamentals. We structurally estimate stock price informativeness for several countries and show that it declines when alternative liquidity sources, such as banks, are in distress. To study the real effects of this mechanism, we devise a strategy to integrate our stock-trading module into a dynamic general equilibrium model with heterogeneous firms. We calibrate the model to the United States and simulate recessions with and without banking distress. In a standalone recession, prices become more informative and allocation improves, mitigating output losses by 4.4%. If the recession coincides with banking distress, agents rely more on stock markets to obtain liquidity, prices become less informative, and allocation deteriorates, magnifying output losses by 22%. *JEL codes:* E44, E32, G14.

I. INTRODUCTION

Stock markets play two critical roles in market economies: allocating resources and providing liquidity. The former stems from the remarkable ability of markets to aggregate and make pub-

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lic information about firms' prospects that is dispersed among traders. This is usually referred to as stock price informativeness, guiding how investors allocate their funds across investment opportunities. The latter, commonly referred to as market liquidity, comes from providing a platform where traders can buy or sell stocks to satisfy their liquidity needs. The extent to which traders rely on stock markets for liquidity depends on the availability of alternative sources, such as financial intermediaries that provide liquidity by facilitating credit, commonly referred to as funding liquidity.

Despite a rich literature discussing the information and liquidity roles of stock markets separately, their interaction is less understood. Does the role of stock markets in providing liquidity enhance or weaken their role in providing information? How does distress in other sources of liquidity, such as the banking sector, affect price informativeness? How does the information content of stock prices vary with business cycles? How do banking crises and other credit market distress contribute to the depth of a recession?

In this article, we make progress on three fronts. The first is theoretical, by building a model of stock trading where the extent to which agents use stocks to access liquidity affects the extent to which prices reflect firms' fundamentals. We incorporate this module into a dynamic general equilibrium setting with heterogeneous firms where stock price informativeness affects capital allocation and investment. The second front is empirical, by structurally estimating stock price informativeness from firm-level panel data for several countries and establishing its cyclical properties. The last front of progress is quantitative, by calibrating the general equilibrium model to quantify the role of price informativeness on capital allocation in recessions, with and without funding liquidity distress.

To be more precise, our theoretical contribution hinges on extending the seminal model of [Grossman and Stiglitz \(1980\)](#): rational traders are willing to acquire information about firms at a cost because the presence of noise traders prevents prices from perfectly revealing the information others acquire. To accommodate the two main roles of stock markets, we allow for two types of rational traders—day and night—interested in different stock properties—liquidity and fundamentals, respectively. This structure dispenses with the assumption of noise traders and creates endogenous noise in prices: a high price may indicate that the stock is easily traded or that the firm has strong fundamentals.

The trades from one type of trader act as noise for the other. We show that, in equilibrium, (i) a linear pricing function exists where price informativeness depends on the relative weights of information about fundamentals and about liquidity contained in prices, and (ii) these weights are given by how many day and night traders operate in the market, their information choices, and how aggressively they trade on their information.

We integrate this trading module into a real business cycle model with heterogeneous firms. We provide a parsimonious link between financial and real markets by preserving a linear pricing function in a nonlinear production economy. Stocks are claims to firms' earnings, which depend on firms' idiosyncratic productivities (i.e., fundamentals). If those productivities were known, investors would allocate capital efficiently across firms. In our model, those productivities are *ex ante* unknown, and allocations are based on investors' best estimates derived from stock prices, which conflate fundamental information with noise generated by day traders. When funding markets malfunction, the economy relies more on stocks to access liquidity, there are more day traders, price informativeness declines, and the misallocation of resources across firms worsens.

Our model delivers a linear relationship among stock prices, firm earnings, and stock liquidity, independent of general equilibrium considerations. This property allows us to define price informativeness (PI) as an analytical function of the dispersion of firm productivity, the dispersion of stock price fluctuations, and their respective price loadings. This measure tells us that in a market with high PI, a firm's high stock price (relative to other firms) is a strong signal that traders have positive information about its fundamentals. Our empirical contribution consists of measuring the PI in a given country each year. We implement this structural estimation using panel data from 16 countries spanning 1984 to 2022 and show that PI exhibits cyclicalities and more importantly declines in periods of insufficient funding liquidity, such as the Great Recession and the COVID-19 pandemic. We structurally decompose PI fluctuations into its four components and show that these fluctuations are mostly explained by changes in trading activity rather than in information quality.

Finally, in our quantitative contribution, we measure the relevance of stock price informativeness on the allocation of resources, total investment, and other macro aggregates. We calibrate the parameters of the full model for the United States, assuming the

economy is subject to two, possibly correlated, aggregate shocks: one affecting aggregate productivity and one affecting funding liquidity. The shock structure is meant to capture recessions with and without distress in the banking sector. We use aggregate moments and moments of the estimated pricing functions to discipline the cost of acquiring information and the dynamics of market liquidity needs. The calibrated model successfully replicates the cyclical properties of our PI measure without explicitly targeting them.

Our PI estimate measures how much prices reveal to an outside observer what informed traders know. In the model, stock prices are the only sources of information about firms, but in reality investors use other sources and may already know what informed traders know. We acknowledge these possibilities in our quantitative analysis, disciplining their relevance using empirical price-investment correlations across firms and time-series correlations between PI and investment dispersion. We show that new stock price information for decision makers is roughly as important as the information they already know, thereby underscoring the “revelatory” importance of PI.

Using the calibration, we conduct impulse-response exercises to simulate recessions with and without distress in funding markets. Recessions without funding market distress are cleansing. Increased uncertainty induces all traders to acquire more information, leading to greater PI and improved resource allocation, mitigating output losses by 4.4%. In contrast, recessions with funding market distress are sully. Despite all traders acquiring more information, the heightened trading activity for liquidity purposes makes prices less informative, and resource allocation worsens, magnifying output losses by 22%. This result implies a sizable real effect of banking problems through a novel channel: weakening the information and allocative role of stock markets.

We further explore how the U.S. economy would fare in a recession with funding liquidity distress under alternative information structures. If information were exogenous, liquidity distress would reduce PI by a larger amount, with output declining by 43% more than in our benchmark with endogenous information. This result suggests that endogenous information acquisition makes stock markets “lean against the wind” during recessions, by conveying more information about fundamentals. We also show that if the cost of information about firms’ fundamentals were halved, PI would decline less, leading to output declines that are about 5%

smaller. If instead the cost of information about a stock's liquidity were halved, PI would decline more, leading to output declines that are about 2% larger. This result suggests that transparency is nuanced: easier access to information about one source can make it harder to infer information about other sources.

Our article lies at the intersection of the literature on PI and the literature on input misallocation. Also on this intersection are [David, Hopenhayn, and Venkateswaran \(2016\)](#) and [David and Venkateswaran \(2019\)](#), perhaps the closest to our study. The former focuses on the role of informational frictions in resource allocation and measures how much each source of information contributes to productivity gaps. The latter has a larger scope and incorporates many potential frictions that can distort resource allocation in addition to informational frictions. Both studies provide static measures; hence, they are silent about cyclicalities, which is our focus. Furthermore, by formalizing how the information content in stock prices reacts to changes in aggregate conditions, we can discipline the relative weight assigned to stock prices when allocating resources over the cycle.¹

Our work addresses seemingly conflicting results in the literature on input allocation across firms during recessions.² We show that those findings should be qualified by whether a recession coincides with distress in financial markets. We provide a mechanism that accounts for [Foster, Grim, and Haltiwanger \(2016\)](#)'s finding that labor allocation worsened during the Great Recession, despite improving during previous post-1980s U.S. recessions. Further, by quantitatively assessing the role of liquidity distress in misallocation, we provide a complementary channel through which financial markets affect business cycles: when

1. [Goldstein \(2023\)](#) provides a survey of the evidence about the relevance of stock prices in informing and affecting the decisions of managers, banks, credit rating agencies, and regulators. Also, even though their focus is not misallocation, [Dow, Goldstein, and Guembel \(2017\)](#), [Benhabib, Liu, and Wang \(2019\)](#), and [Chousakos, Gorton, and Ordoñez \(2023\)](#) provide theoretical interactions between real and financial sectors in terms of learning and information acquisition.

2. [Alam \(2020\)](#) documents a countercyclical dispersion of the marginal product of capital (MPK) using balance sheet data from North America and Europe. See also [Oberfield \(2013\)](#) and [Sandleris and Wright \(2014\)](#), which document an increased dispersion of MPK during the 1982 Chilean crisis and the 2001 Argentine crisis, respectively. [Flynn and Sastry \(2024\)](#) and [Osotimehin \(2019\)](#) claim the opposite: the former argues that US public firms' input choices become more careless and volatile, and the latter argues that within-sector allocative efficiency among French firms declines in good times.

banks are in distress, the informational and allocative roles of stock markets weaken. Hence, we add to the important literature that follows [Bernanke \(1983\)](#) on understanding how financial problems magnify fluctuations.

We contribute to the theoretical and empirical literature on PI. On the theoretical front, the literature usually assumes an exogenous source of noise that prevents prices from being perfectly informative. A follow-up literature, such as [Stein \(1987\)](#) and [Vives \(2014\)](#), endogenizes the information:noise ratio by assuming that two dimensions of information are condensed into a single price. Both use heterogeneity in traders' characteristics (the former on market access and the latter on private valuations) to generate imperfectly informative prices without exogenous noise.³ We also endogenize noise in prices, but heterogeneous traders acquire common, not private, information about two dimensions of the stocks. In other words, instead of traders with access to asymmetric information, we have traders with asymmetric needs.

On the empirical front, we contribute to a recent literature that strives to measure stock PI. [Dávila and Parlatore \(2026\)](#) use time-series regressions to measure a moving-average price informativeness for each stock, which requires assumptions on how parameters change over time to keep the cross-sectional variation flexible. We use cross-sectional regressions to measure the price informativeness of a representative stock over time, which requires us to make assumptions on the extent of heterogeneity across stocks to allow parameters to change flexibly over time. [Bai, Philippon, and Savov \(2016\)](#) also resort to cross-sectional regressions but estimate a measure based on the forecasting ability of prices and focus on its long-run trends, not its cyclical properties. Our PI measures how well stock prices reflect what informed traders know at a given time, which is related to but distinct from their ability to forecast future returns. Our measure is also silent on whether such information is incorporated into decision making, but in the quantitative analysis, we devise a method to identify its "revelatory" component.⁴

3. See also [Rahi \(2021\)](#) and [Banerjee, Davis, and Gondhi \(forthcoming\)](#) for theoretical exercises that expand on [Vives \(2014\)](#).

4. Our estimation strategy distinguishes "forecasting" and "revelatory" PI, as defined in [Bond, Edmans, and Goldstein \(2012\)](#). While the former concept captures how well prices predict future earnings, and affect what [Dow and Gorton \(1997\)](#) call "stock market efficiency," the latter captures how well prices inform decision makers and real variables, being relevant for what they call "economic efficiency."

The article proceeds as follows. [Section II](#) introduces a stock-trading model with endogenous noise and shows how to integrate it into an otherwise standard real business cycle model with firm heterogeneity. [Section III](#) describes the data sources and the empirical strategy to compute stock PI and its structural components. Then, we study the cyclical properties of the estimated series and their relation with measures of funding liquidity distress. [Section IV](#) calibrates the model to the United States. [Section V](#) uses the calibrated model to assess the quantitative relevance of liquidity distress, information costs, and additional sources of information for real variables through their effects on PI. [Section VI](#) concludes. All proofs are in [Online Appendix A](#).

II. MODEL

We build a stock-trading model and incorporate it into a real business cycle model with firm heterogeneity. A firm's stock price provides information about two dimensions: the firm's profitability and the stock's liquidity, which are valued differently by different types of traders. A single price contains information about both elements; the trading activity of one type of trader masks the information from the other. Because real investors use a firm's stock price as a signal of profitability, allocative efficiency depends on the composition of traders and their information-acquisition choices. To keep the notation simple, we suppress time subscripts unless necessary.

II.A. Environment

1. *Preferences.* Time is discrete. In a given period, the economy is populated by a measure one of identical infinitely lived households and a measure one of traders who live for just one period. The representative household has constant relative risk aversion (CRRA) preferences where the utility from consuming W is given by $u(W) = \frac{W^{1-\eta}-1}{1-\eta}$, with intertemporal elasticity of substitution $\frac{1}{\eta}$. Traders have constant absolute risk aversion (CARA) preferences where the utility from consuming W is given by $v(W) = -e^{-aW}$, with risk aversion $a > 0$,⁵ indexed by $t = 0, 1, 2, \dots$

5. We can allow long-lived traders. With CARA utility functions, their portfolio choice problem would not be affected by wealth, and with our assumptions of i.i.d. shocks over time, the solution would be identical. We could also allow households

2. *Technology.* There is a measure one of firms (indexed by i) with profit function:

$$(1) \quad \Pi_i = z_{in}(\bar{K}_i + K_i) - \xi \frac{K_i^2}{2\bar{K}_i} - r_i K_i,$$

where z_{in} is the productivity of firm i 's capital, $\bar{K}_i$ is installed capital, and K_i is capital rented at a price r_i . While $\bar{K}_i$ cannot be changed or reallocated, K_i is chosen every period. This profit function is derived from a production function with capital and a quadratic adjustment cost, which introduces curvature. The output price is normalized to one.

3. *Assets and Endowments.* There are three types of assets in the economy: capital, stocks, and foreign bonds. Households save in capital through a mutual fund. The mutual fund allocates capital by renting it to firms through price-discriminating, take-it-or-leave-it offers $\{K_i, r_i(z_{in})\}$ to each firm i . In words, the mutual fund rents K_i to firm i charging a rate $r_i(z_{in})$ that is conditional on the realized ex post productivity, extracting all surplus in each state.⁶ The mutual fund is a decision-making aggregator owned by households, which is meant to capture households' allocation of their savings across different firms. Hence, the mutual fund's total return r is fully distributed to households. Traders can invest in foreign bonds with exogenous return r^F and in firms' shares, that is, stocks. Each share entitles the holder to one unit of installed capital at the associated firm, so the firm's outstanding share amount, i , equals $\bar{K}_i$.⁷

This market structure provides a clear split of the firm surplus: the surplus from $\bar{K}_i$ goes to traders, while the surplus from K_i goes to households through the mutual fund. This split guar-

to be traders, but our modeling choice provides a cleaner exposition by separating the aggregation of information from the use of information.

6. Note that although K_i cannot be conditioned on z_{in} , the interest rate can be. The conditional interest rate implies that (i) while capital needs to be assigned before z_{in} is observed, the exact payment is determined later to extract all the ex post surplus, and (ii) there is no default.

7. The foreign bonds prevent the interest rates determined in general equilibrium from feeding back into prices. Relaxing this assumption would require estimating our proposed PI measure in a general equilibrium setting, thereby breaking the clean separation between informativeness and allocations. As we discuss later, the interest rate affects only the intercept of the pricing function and is irrelevant for the price's signaling ability and for our purposes.

antees that the profitability of holding a firm's stock depends only on z_{in} , not on the capital K_i allocated to the firm.

4. Funding Liquidity and Market Liquidity. Traders are born at the beginning of the period, endowed with $\tilde{b}$ foreign bonds. They can buy stocks at birth (from previous traders about to die) and sell stocks at the end of the period (to newborns) but need to consume in the middle of the period. They have two ways to finance such consumption. They can obtain a loan from a bank (funding liquidity) or sell the stocks in advance (market liquidity).⁸

We assume a fraction γ of traders do not have access to funding liquidity. They are forced to sell their stocks “during the day” at a liquidity discount z_{id} , which is stock-specific. We call these traders relying on market liquidity day traders. In contrast, the rest of the traders can obtain a loan at no cost (just a normalization) and hold stocks until selling “at the end of the day,” so we call them night traders. Everybody observes γ , and each trader knows its type at birth. Changes in γ are then meant to capture changes in funding liquidity: an increase in γ reflects distress in banks and/or the bond market. Changes in the liquidity discount z_{id} are meant to capture changes in a given stock's market liquidity, defined as the losses a day trader faces from selling the stock in the middle of the period (a liquidity discount). In what follows, we model z_{id} as exogenous, but in [Online Appendix C](#), we endogenize it as the fee a day trader has to pay a dealer to compensate for the risk involved in buying the stock, holding it, and selling it later at a different price.

5. Information. A firm's stock is characterized by two dimensions in each period, the firm's profitability z_{in} and the stock's liquidity discount z_{id} .

The firm's profitability has three components: an aggregate productivity shock (Z), a random term that can be learned (θ_{in}),

8. This assumption of infinite marginal utility to consume intraperiod is meant to capture stressful situations in which the marginal utility of obtaining funds intraperiod (e.g., margin calls or covering impending debts due to payment failures) is sufficiently high. The relevance of stocks for liquidity, not only for financial institutions but also for high-income households after income shocks, was recently documented by [Adams \(2026\)](#). Even though we focus on banks as the main providers of funding liquidity, bond markets also play this role, underscoring the importance of market liquidity during bond market disruptions.

and a random term that cannot be learned (ε_{in}). The learnable component θ_{in} is drawn from a prior distribution $\mathcal{N}(\bar{\theta}_{in}, \sigma_{\theta_{in}}^2)$, while the unlearnable component $\tilde{\varepsilon}_{in}$ follows an AR(1) process: $\tilde{\varepsilon}_{in} = \rho \tilde{\varepsilon}_{in}^- + \varepsilon_{in}$ with $\varepsilon_{in} \sim \mathcal{N}(0, \sigma_{\varepsilon_{in}}^2)$, where $\tilde{\varepsilon}_{in}^-$ is public information and $\sigma_{\varepsilon_{in}}^2$ is a measure of fundamental uncertainty. The stock's liquidity discount (z_{id}) has two components: a random term that can be learned (θ_{id}) and a random term that cannot be learned (ε_{id}). The first is drawn from $\mathcal{N}(\bar{\theta}_{id}, \sigma_{\theta_{id}}^2)$ and the second from $\mathcal{N}(0, \sigma_{\varepsilon_{id}}^2)$.⁹

To summarize, the firm's profitability and the stock's liquidity discount are

$$\begin{aligned} z_{in} &= Z + \theta_{in} + \tilde{\varepsilon}_{in}, \\ \text{where } \theta_{in} &\sim \mathcal{N}(\bar{\theta}_{in}, \sigma_{\theta_{in}}^2), \\ \tilde{\varepsilon}_{in} &= \rho \tilde{\varepsilon}_{in}^- + \varepsilon_{in}, \quad \& \quad \varepsilon_{in} \sim \mathcal{N}(0, \sigma_{\varepsilon_{in}}^2), \\ z_{id} &= \theta_{id} + \varepsilon_{id}, \\ (2) \quad \text{where } \theta_{id} &\sim \mathcal{N}(\bar{\theta}_{id}, \sigma_{\theta_{id}}^2) \quad \& \quad \varepsilon_{id} \sim \mathcal{N}(0, \sigma_{\varepsilon_{id}}^2), \end{aligned}$$

where θ_{id} , θ_{in} , ε_{id} , and ε_{in} are independently distributed across firms and over time. We allow $\sigma_{\theta_{in}}^2$, $\sigma_{\theta_{id}}^2$, $\sigma_{\varepsilon_{in}}^2$, and $\sigma_{\varepsilon_{id}}^2$ to be functions of aggregate productivity Z , but not of γ . This way, we guarantee that liquidity distress would not affect economic outcomes if it were not for changes in the information content of prices.

Traders can perfectly learn both θ_{id} and θ_{in} by paying a cost.¹⁰ The fraction of $l \in \{d, n\}$ traders who choose to be informed about stock i is denoted with λ_{il} . The information cost may be different for night traders $c_n(\lambda_{in})$ and for day traders $c_d(\lambda_{id})$, and potentially depend on the fraction of informed traders. The unlearnable components $\varepsilon = \{\varepsilon_{id}, \varepsilon_{in}\}_{i=0}^1$ can be freely observed only after their realization. Finally, the mutual fund doesn't have access to this information technology and, like uninformed traders, infers z_{in} solely from observing stock market prices. We relax this last assumption when calibrating the model, allowing the mutual fund to obtain independent signals about θ_{in} and ε_{in} .

9. The liquidity discount z_{id} could also depend on γ . As long as it affects all the firms additively, in the same way that Z affects z_{in} , our analysis remains unchanged. Also, it is straightforward to allow persistence in ε_{id} , as we assume for ε_{in} , but empirically its role is negligible (see [Online Appendix D.2](#)).

10. That traders learn about both signals is without loss of generality, as in equilibrium the trader could combine one signal with the price to infer the other signal perfectly.

The aggregate productivity Z and the fraction of day traders γ define a state $s = \{Z, \gamma\}$, which is public information, and follow an exogenous Markov process with joint transition probability $q_{s,s'}$.

6. *Timing.* Each period starts with the realization of γ , Z and $\theta = \{\theta_{in}, \theta_{id}\}_{i=0}^1$. Then:

- i. Day and night traders simultaneously choose whether to acquire information and how many stocks to buy from each firm.
- ii. Households make saving decisions and invest in the mutual fund, which allocates capital across firms after observing all stock prices.
- iii. $\varepsilon = \{\varepsilon_{in}, \varepsilon_{id}\}_{i=0}^1$ is realized, with both θ and ε becoming public information.
- iv. Day traders sell all their stocks at a discount z_{id} , consume, and die.
- v. Production occurs, firms pay the mutual fund, and the fund pays households.
- vi. Night traders sell their stocks, consume, and die.

II.B. Traders' Problems and Stock Market Clearing

1. *Portfolio Choices.* A trader of type $l \in \{d, n\}$ chooses a portfolio of foreign bonds and stocks from each firm:

$$(3) \quad \max_{B_l, \{X_{il}\}_{i \in [0,1]}} E \left[-\exp \left[-\alpha \left[(1 + r^F) B_l + \int_i X_{il} (z_{in} - \mathbb{I}_{\{l=d\}} z_{id} + p'_i - p_i) di \right] \right] \right]$$

$$s.t. \quad B_l + \int_i p_i X_{il} di = \tilde{b},$$

where $\mathbb{I}_{\{l=d\}}$ is the indicator function for a day trader, and B_l and $\{X_{il}\}_{i \in [0,1]}$ denote the demands for foreign bonds and stocks. This is a standard portfolio rebalancing problem between safe assets (foreign bonds at a risk-free rate of r^F) and risky assets (stocks).

Given the distributional assumptions and the information structure (a standard CARA-normal setting), the stock demand

functions are :

$$\begin{aligned}
 X_{in}^{I*} &= \frac{E[z_{in} + p'_i|\theta] - (1 + r^F)p_i}{a\text{Var}[z_{in} + p'_i|\theta]}, \\
 X_{id}^{I*} &= \frac{E[z_{in} - z_{id} + p'_i|\theta] - (1 + r^F)p_i}{a\text{Var}[z_{in} - z_{id} + p'_i|\theta]}, \\
 X_{in}^{U*} &= \frac{E[z_{in} + p'_i|p] - (1 + r^F)p_i}{a\text{Var}[z_{in} + p'_i|p]}, \\
 (4) \quad X_{id}^{U*} &= \frac{E[z_{in} - z_{id} + p'_i|p] - (1 + r^F)p_i}{a\text{Var}[z_{in} - z_{id} + p'_i|p]},
 \end{aligned}$$

where X_{in}^{I*} , X_{id}^{I*} , X_{in}^{U*} , and X_{id}^{U*} represent the stock i 's demand by night and day traders that are informed (I) and uninformed (U), respectively. While the informed condition expectations and variances on θ , the uninformed do it on p . The difference between the demand of day and night traders is that day traders' payoffs depend on both z_{in} and z_{id} .

2. Information Acquisition Choices. Denoting the information acquisition choice of trader j about firm i as $I_{ji} \in \{0, 1\}$, the end-of-period wealth of trader j of type $l \in \{d, n\}$ is

$$\begin{aligned}
 W_{lj}(\mathbf{I}_j) &= (1 + r^F) \left(W_{0j} - \int_i I_{ji} c_l(\lambda_{il}) di \right) \\
 (5) \quad &+ \int_i (z_{in} - \mathbb{I}_{l=d} z_{id} + p' - (1 + r^F)p_i) (I_{ji} X_{il}^{I*} + (1 - I_{ji}) X_{il}^{U*}) di,
 \end{aligned}$$

where $\mathbf{I}_j$ is the vector of information choices of agent j for all stocks. Let $\mathbf{I}_j^0$ be an arbitrary information acquisition vector where the element I_{ji}^0 for a specific stock i is zero and $\mathbf{I}_j^1$ be an identical vector, except that I_{ji}^1 is one. The trader would choose $\mathbf{I}_j^1$ over $\mathbf{I}_j^0$, that is, fixing information about all stocks, a trader j acquires information about stock i , if and only if $E \left[v \left(W_{lj}(\mathbf{I}_j^1) \right) \mid \mathbf{p} \right] \leq E \left[v \left(W_{lj}(\mathbf{I}_j^0) \right) \mid \mathbf{p} \right]$ where $\mathbf{p}$ is the vector of stock prices. In words, the trader would like to acquire information if it reduces the expected variance of her end-of-period wealth, net of the information cost. Formally, we have **Lemma 1**.

LEMMA 1. Let $\mathbf{I}_j^0$ be an arbitrary information acquisition vector where $I_{ji}^0 = 0$ for a specific stock i and $\mathbf{I}_j^1$ be an identical vector,

except that $I_{ji}^1 = 1$. Defining λ as the set of λ_{il} for all firms and traders of type $l \in \{d, n\}$, a trader j of type l would acquire information about firm i conditional on others' information choices λ if and only if,

$$\frac{E \left[v \left(W_{l,j}(\mathbf{I}_j^1) \right) \mid \mathbf{p} \right]}{E \left[v \left(W_{l,j}(\mathbf{I}_j^0) \right) \mid \mathbf{p} \right]} = e^{ac_l(\lambda_{il})} \sqrt{\frac{\text{Var}[z_{in} - \mathbb{I}_{\{l=d\}} z_{id} + p'_i | \theta_i]}{\text{Var}[z_{in} - \mathbb{I}_{\{l=d\}} z_{id} + p'_i | p_i]}}$$

(6) $\equiv \psi^{il}(\lambda) \leq 1.$

Equation (6) shows the ratio of the expected variance of end-of-period wealth between acquiring information about stock i or not, all else equal. While the first term captures the cost of acquiring information about firm i and is greater than one, the second term captures the benefit of reducing variance by observing θ_i rather than just p_i and is less than one. Because traders want to reduce the variance of payoffs, they would acquire information if and only if this ratio, defined as $\psi^{il}(\lambda)$, is less than one. The combination of CARA-normal properties, independent returns across stocks, and independent information-acquisition costs guarantees that whether to acquire information about one stock is independent of whether to acquire information about another.

Corollary 1 shows equilibrium conditions for λ_{il} ,

COROLLARY 1. When $\sigma_{\theta_d}^2 > \sigma_{\theta_n}^2$, $\psi^{il}(\lambda)$ is monotone in λ_{il} for $l \in \{d, n\}$. Therefore:

- i. If $\psi^{il}(\lambda) < 1 \ \forall \lambda_{il} \in [0, 1]$, all l traders become informed about stock i , that is, $\lambda_{il}^* = 1$.
- ii. If $\psi^{il}(\lambda) > 1 \ \forall \lambda_{il} \in [0, 1]$, no l traders become informed about stock i , that is, $\lambda_{il}^* = 0$.
- iii. Otherwise, λ_{il}^* is given by $\psi^{il}(\lambda_{il}^*) = 1$.

3. *Stock Market Clearing.* Market clearing for the shares of firm i 's installed capital is

$$(7) \quad \gamma \left[\lambda_{id} X_{id}^I + (1 - \lambda_{id}) X_{id}^U \right] + (1 - \gamma) \left[\lambda_{in} X_{in}^I + (1 - \lambda_{in}) X_{in}^U \right] = \bar{K}_i.$$

II.C. Real Sector Problems and Capital Market Clearing

1. *The Representative Household's Problem.* Its recursive formulation is :

$$(8) \quad H(s, K, k) = \max_{k'} u(k(1 + r(s, K) - \delta) - k') + \beta \sum_{s'} q_{ss'} H(s', K', k')$$

$$s.t. \quad K' = G(K),$$

where $s = \{Z, \gamma\}$ denotes the aggregate state, $H(\cdot)$ the value function, β the discount factor, δ the depreciation rate, k the individual's capital holdings, and K the aggregate capital holdings. $G(\cdot)$ represents household expectations over the future path of aggregate capital. Importantly, households decide on savings in capital given the interest rate r that mutual funds pay, which depends on how well the mutual fund allocates capital.

2. *The Mutual Fund's Capital Allocation Across Firms.* We model the mutual fund as an aggregator that collects households' savings, allocates capital to maximize total surplus, and distributes the surplus back to households with perfect diversification. Since we assume the mutual fund can extract all ex post surplus from the rented capital and pass it to households, it allocates K_i to firm i to maximize firm i 's expected profits conditional on observed prices. Given the cost of funding r and adjustment costs, the capital allocated to firm i is given by

$$(9) \quad K_i = \max \left\{ \bar{K}_i \left(\frac{E[z_{in} | \mathbf{p}] - r}{\xi} \right), 0 \right\} \quad \forall i,$$

and is implemented by a take-it-or-leave-it offer with a contract $\{K_i, r_i(z_{in})\}$ that satisfies

$$(10) \quad z_{in} K_i - r_i(z_{in}) K_i - \xi \frac{K_i^2}{2\bar{K}_i} = 0 \quad \forall i, z_{in} > 0.$$

Hence, the mutual fund captures all ex post surplus from the rented capital, while traders capture the ex post surplus from the installed capital. The mutual fund distributes all surplus to households, such that

$$\int_i r_i K_i = rK,$$

where K is the total capital supplied by households.

In an environment where θ_{in} is observable, the fund would choose K_i based on $E[z_{in} | \theta_{in}]$. Hence, "misallocation" would be due

to ε_{in} , which is inevitable. In our setting, however, the mutual fund can only rely on stock prices to infer θ_{in} , hence using $E[z_{in}|\mathbf{p}]$ to make decisions. Given idiosyncratic shocks are i.i.d., the only price that contains information about θ_{in} is p_i . For the choice of K_i , θ_{id} is irrelevant, yet a high stock price could stem from a high θ_{in} or a low θ_{id} . In summary, the presence of day traders prevents prices from perfectly revealing θ_{in} to the mutual fund and worsens capital allocation.

3. *Capital Market Clearing.* Non-installed capital market clears,

$$(11) \quad \int_i K_i = K.$$

II.D. Equilibrium

DEFINITION 1. $H, r, k', G, \{K_i, r_i, X_{id}^I, X_{id}^U, X_{in}^I, X_{in}^U, \lambda_{id}, \lambda_{in}, \phi_{i0}, \phi_{i\varepsilon}, \phi_{id}, \phi_{in}, p_i\}_{i \in (0,1)}$ constitute a linear rational expectations equilibrium (LREE) such that

- i. $X_{id}^I, X_{id}^U, X_{in}^I$, and X_{in}^U solve traders' portfolio choices [equation \(3\)](#), characterized in [equation \(4\)](#).
- ii. $\lambda_{id}, \lambda_{in}$ solve traders' information acquisition problems, characterized by [Lemma 1](#).
- iii. The stock price of firm i is a linear function of θ_{id} and θ_{in} , that is, $p_i = \phi_{i0} + \phi_{i\varepsilon}\varepsilon_{in}^- + \phi_{id}\theta_{id} + \phi_{in}\theta_{in}$, where $\phi_{i0}, \phi_{i\varepsilon}, \phi_{id}$, and ϕ_{in} solve stock market clearing [equation \(7\)](#).
- iv. k' and H solve the representative consumer's problem in [equation \(8\)](#).
- v. K_i, r_i solve the mutual fund's problem in [equations \(9\) and \(10\)](#).
- vi. r clears non-installed capital market [equation \(11\)](#).
- vii. G is consistent with k' .

Notice we restrict attention to LREE. Linear pricing is needed to preserve Gaussian conjugates so the agents' Bayesian updating remains analytical. If the price is nonlinear, this update must be performed via numerical integration, rendering the model a large black box. In this case, we would not be able to characterize demand, derive a price equation, or estimate a structural measure of price informativeness. Before presenting these results, we high-

light the assumptions that make them possible. Unfortunately, we cannot directly test these assumptions, as relaxing them makes the model intractable.

1. *Assumptions Relevant for an LREE.* Besides the standard CARA-normal assumptions, we resort to additional structures to preserve linear pricing.

One set of assumptions relates to the distribution of capital surplus in the economy. First, production is additive in fixed $\bar{K}$ (tradable) and adjustable K (investments). Second, firms rent K , which does not accumulate within a firm. Third, the mutual fund extracts all surplus from renting K to firms and passes it to households. These assumptions break the potential feedback loop from capital markets to stock markets: if firms retained surplus from their investments, this surplus would go to traders. Then traders recognize that higher prices attract greater investment, which in turn increases demand and raises prices further. Because the allocated capital is nonlinear in productivity, the price becomes a nonlinear function of the fundamentals.¹¹ [Albagli, Hellwig, and Tsyvinski \(2023\)](#) theoretically show that such feedback can be strong when investors have incentives to manipulate stock prices. Although in our setup the mutual fund is the sole determinant of investment, in reality, many decision makers (managers, boards, banks, etc.) share this role. Hence, coordinating stock price manipulation is likely more difficult.

Second, we assume that persistent differences across firms' productivity are given by the non-learnable component, $\tilde{\varepsilon}_{in}$, which follows an AR1 process. This assumption renders the resale price unlearnable, preserving the current price linearity. [Farboodi and Veldkamp \(2020\)](#) show that persistency of the learnable component, θ_{in} , also preserves price linearity but when traders are homogeneous. In our empirical analysis, we indirectly relax this assumption by residualizing prices using previous earnings as a proxy for $\bar{\theta}_{in}$.¹²

11. [Subrahmanyam and Titman \(1999\)](#) make similar assumptions and interpret them as a world with derivative assets providing claims on a firm's cash flows rather than on all of its assets. Because traders can back out the capital allocated to a firm, the amount of information revealed does not depend on whether the claim is on cash flows or all assets.

12. Albeit implicitly, we also assume that *firms do not use prices to incentivize managers*, a possibility explored in [Banerjee, Davis, and Gondhi \(forthcoming\)](#). We show in [Online Appendix D.1](#) that in a stylized extension where the firm's pro-

II.E. Price Informativeness

1. Pricing Function. We show that conditional on the fraction of informed traders, there exists a unique linear pricing function (Proposition 1). Then we characterize the linear rational expectations equilibrium that arises when information is endogenous.

PROPOSITION 1. Given a fraction of informed traders, (λ_{in} and λ_{id}), there exists a unique linear market price for the stock of firm i that adopts the form $p_i = \phi_{i0} + \phi_{i\varepsilon}\varepsilon_{in}^- + \phi_{id}\theta_{id} + \phi_{in}\theta_{in}$, where $\phi_{in} > 0$, $\phi_{id} < 0$, and

$$(12) \quad \frac{|\phi_{in}|}{|\phi_{id}|} = 1 + \frac{(1 - \gamma)\lambda_{in}(\sigma_{\varepsilon_{id}}^2 + \text{Var}(z_{in} + p'_{in}))}{\gamma\lambda_{id}\text{Var}(z_{in} + p'_{in})}.$$

The ratio $\frac{|\phi_{in}|}{|\phi_{id}|}$ in equation (12) captures the impact of θ_{in} on the price relative to θ_{id} , hence how much can be learned about the learnable component of firms' fundamentals θ_{in} from observing p_i . Although endogenous information acquisition is relevant for quantitatively disciplining the extent of price informativeness, it is not critical qualitatively for the mechanism. Note that even if all traders are informed (i.e., $\lambda_{in} = \lambda_{id} = 1$), price informativeness depends on the fraction of day traders, γ . In the extremes, as the fraction of day traders approaches zero, the ratio would approach infinity, and prices would perfectly reveal what traders know about fundamentals. In contrast, as the fraction of day traders approaches one, the ratio approaches one, and prices would have the same loadings on fundamentals and liquidity, given that day traders care about both. Corollary 2 provides in words comparative statics about the information about θ_{in} contained in p_i .

COROLLARY 2. *Ceteris paribus*, price becomes less sensitive (i.e., informative) about θ_{in} when

- i. a larger fraction of traders are day traders,
- ii. a larger fraction of day traders are informed compared to night traders,
- iii. the liquidity discount has a smaller residual variance on information than productivity.

ductivity is linear in managerial effort and managerial compensation is linear in prices, incentive provision is orthogonal to the other roles prices serve, and our linear pricing function can also be preserved.

Conditional on the fraction of day and night traders who are informed, linear pricing is unique because our setting does not exhibit complementarities in trading: informed traders do not learn from prices, hence, they do not condition their trading strategies on the information that prices convey. The characterization of the LREE is completed by endogenizing the fraction of traders who are informed, given information-acquisition choices in Lemma 1. These choices are made by night and day traders and depend on the cost of information for each group. An information cost function that is increasing in the fraction of traders who are informed rules out complementarities in information acquisition within a trader type.¹³ Complementarities in information acquisition across trader types are a more fundamental feature: if more day traders acquire information (high λ_{id}), prices are less informative about fundamentals, magnifying the incentives for night traders to acquire information (high λ_{in}). Whether this complementarity translates into multiplicity in LREE depends on the curvature of information costs. Given our assumptions on information costs and across our various calibrations, we did not detect multiple equilibria. Last, any LREE becomes interior, that is, $\lambda_{il} \in (0, 1)$, when $\lim_{\lambda_{il} \rightarrow 0} c_l(\lambda_{il}) = 0$ and $\lim_{\lambda_{il} \rightarrow 1} c_l(\lambda_{il}) = \bar{C}$ where $\bar{C}$ is large enough.¹⁴

2. PI Measure. We define an intuitive measure of price informativeness that can be structurally estimated from the data for many countries. To do this, we introduce a simplification that we maintain.

13. Increasing information costs is a property, for instance, of an upward-sloping supply curve for information or by heterogeneous information acquisition costs across traders, with the traders with the lowest cost acquiring the information first.

14. This is in contrast to other models that replace noise trading with asymmetric information about multiple dimensions, such as Manzano and Vives (2011). In those models, informed traders are ex ante homogeneous and have private signals. When prices are more informative, they rely more on their private signals to trade, making prices even more informative; hence, when a trader relies heavily on private information, others do as well. This gives rise to complementarities in trading and possibly, equilibrium multiplicity. Complementarities in information acquisition can also arise in settings with dispersed information (Ganguli and Yang 2009) and when agents can learn about other agents' beliefs (Banerjee, Davis, and Gondhi 2018). These additional sources of complementarities are absent in our setting.

ASSUMPTION 1. The parameters $\bar{K}_i, \bar{\theta}_{in}, \bar{\theta}_{id}, \sigma_{\varepsilon_{in}}^2, \sigma_{\varepsilon_{id}}^2, \sigma_{\theta_{in}}^2$, and $\sigma_{\theta_{id}}^2$ are firm invariant.

Assumption 1 guarantees that equilibrium fractions of informed investors $\lambda_{in}, \lambda_{id}$ and pricing function parameters $\phi_{i0}, \phi_{i\varepsilon}, \phi_{in}, \phi_{id}$ are also firm invariant. This assumption implies that the only sources of heterogeneity across firms are $\theta_{in}, \theta_{id}, \varepsilon_{in}$, and ε_{id} . In other words, firms independently draw z_{in} and z_{id} from identical distributions, allowing us to exploit the cross-sectional variation in prices to estimate price informativeness.

We define PI as the reduction in the uncertainty about a firm's productivity conditional on observing the price, normalized by the reduction in the uncertainty conditional on observing the learnable component after acquiring it at a cost. In the same spirit as Goldstein, Li, and Yang (2014), this measure does not capture how much stock prices inform about future earnings, but instead how much they inform about what informed traders know.

DEFINITION 2. The PI measure for firm i is defined as :

$$PI_i = \frac{Var[z_{in}] - Var[z_{in}|p_i]}{Var[z_{in}] - Var[z_{in}|\theta_{in}]},$$

where $Var[z_{in}]$ denotes the unconditional variance of z_{in} .

An advantage of this measure is that it lies between zero and one. A PI measure close to zero would indicate that the uncertainty reduction from observing the price is negligible compared to the reduction from acquiring information. A PI measure close to one would suggest that observing the price is almost as useful as acquiring information and observing θ_{in} .

COROLLARY 3. Under Assumption 1, PI is firm invariant and equals

$$(13) \quad PI = \frac{1}{1 + \frac{\sigma_{\theta_{id}}^2}{\sigma_{\theta_{in}}^2} \left(\frac{\phi_d}{\phi_n} \right)^2}.$$

Hence, under Assumption 1, the PI measure can be summarized by two parameters ($\sigma_{\theta_n}^2$ and $\sigma_{\theta_d}^2$) that captures the quality of information about the two components and two equilibrium objects (ϕ_n and ϕ_d) that capture the price effect of such information. Our empirical strategy estimates these four objects using a cross section of firms in a given country to calculate PI annually.

III. MEASURING PRICE INFORMATIVENESS

This section provides a structural estimate of PI for 16 countries, each for an average of 21 years, and examines its cyclical properties.

III.A. Empirical Strategy to Measure PI

Our PI measure can be directly obtained from estimating the four components in [equation \(13\)](#): the variances of the learnable part of stock profitability and stock liquidity ($\sigma_{\theta_n}^2$ and $\sigma_{\theta_d}^2$), and their price loadings (ϕ_n and ϕ_d). We can measure PI without solving for the full equilibrium, because pricing depends only on the behavior of short-lived traders and is independent of investment choices by long-lived households, and without specifying the traders' information acquisition technology. The challenge is to identify series for the signals θ_n and θ_d and to estimate the price loadings ϕ_n and ϕ_d from running a cross-firm stock-pricing regression for each country-year pair.

In taking the model to the data, we have to consider a couple of discrepancies. First, to be consistent with the model and maintain comparability across stocks, we transform all values (e.g., prices and earnings) from per share to per unit of asset (see [Online Appendix B.3.3](#) for details). Second, we formulate the observed prices as a composite of our theoretical prices (in which the demand for stocks only takes into account firms' fundamentals and stocks' liquidity properties) and other variables not included in our model (such as speculation motives, shocks to the supply of a firm's stocks, etc.). Formally, a firm i 's observed prices in a given year are given by

$$p_i^s = p_i + v_i,$$

where p_i is our theoretical construct and v_i are additional factors we do not include in our model. Hence, we run the following cross-firm pricing regression per country-year.

$$(14) \quad p_i^s = \phi_0 + \phi_\varepsilon \varepsilon_{in}^- + \phi_d \theta_{id} + \phi_n \theta_{in} + v_i.$$

The identification of the price loadings on the signals of profitability (ϕ_n) and liquidity (ϕ_d) per country-year is based on a cross-sectional comparison. ϕ_n is pinned down by the extent to which firms with high prices have higher profitability signals. Similarly, ϕ_d is pinned down by the extent to which firms with high prices have smaller liquidity discounts. To measure p_i^s , we use World-

scope from Thomson/Refinitiv for data on stock prices.¹⁵ Consistent coverage began around 1985 in the United States and by 2000 in most economies in our sample.

To run these regressions, we need the signals of profitability and liquidity observed by informed investors, θ_{id} and θ_{in} . A common practice in the literature is to use realized values, that is, z_{id} and z_{in} , as proxies for θ_{id} and θ_{in} . However, according to our model, regressing the price p_i^s on the realized values would lead to biased estimates of the price loadings because the measurement error would be correlated with the realized earnings, as formally shown in [Online Appendix B.1](#). Given this drawback, we need to rely on proposing a yearly estimator of the signals about profitability (that we denote as $\hat{\theta}_{in}$) and about liquidity (that we denote as $\hat{\theta}_{id}$).

1. Signals About Profitability, $\hat{\theta}_{in}$. From [equation \(2\)](#) we know informed traders form expectations about each firm's profitability as follows:

$$E[z_{in}|\theta_{in}] = Z + \theta_{in} + \rho\tilde{\varepsilon}_{in}^-.$$

We can measure $E[z_{in}|\theta_{in}]$ by the median one-step-ahead analyst forecast for z_{in} of company i , which we denote as F_i . To do this, we use the Institutional Brokers Estimate System (I/B/E/S) from Refinitiv for daily data on analyst forecasts of earnings per share.¹⁶ We access Worldscope and I/B/E/S through Wharton Research Data Services (WRDS), which provides a unique ticker for each company to link the two data sets. In the [Online Appendix](#),

15. Worldscope is a leading source of cross-country financials of publicly traded companies. It provides data from 98,000 companies in more than 120 countries, which amounts to 99% of the global market capitalization in 2021. Worldscope uses a standardized template for financial information that corrects for measurable differences in accounting practices across companies and markets. The entries are subject to automated tests that verify accounting identities, detect outliers, and assess correlations for accuracy. See Reuters' Worldscope Definitions for details on standardization practices, accuracy tests, coverage, and sample selection criteria.

16. I/B/E/S collects forecasts about 22,000 active companies in 90 countries from over 18,000 analysts. Each observation is an analyst's forecast announcement regarding a company's balance sheet item for a particular horizon. Forecasts are available for various payoff-relevant items, but earnings per share (EPS) has the widest coverage. We use forecasts announced within a 30-day window around the date the stock price is documented.

we discuss the steps we took to make firms and countries comparable.¹⁷

We compute Z by the cross-sectional average of F_i . To decompose θ_{in} from $\rho\tilde{\varepsilon}_{in}$, we construct the time series of forecast errors $\varepsilon_{in} = z_{in} - F_i$ for firm i and compute its variance $\sigma_{\varepsilon_n}^2$. Then we obtain ρ from the autocovariance of the deviation of F_i from its cross-sectional average $E[F_i]$, since

$$\begin{aligned} & \text{Cov}(F_{it} - E[F_{it}], F_{i,t+1} - E[F_{i,t+1}]) \\ &= \text{Cov}(\theta_{int} + \rho\tilde{\varepsilon}_{in,t-1}, \theta_{in,t+1} + \rho(\rho\tilde{\varepsilon}_{in,t-1} + \tilde{\varepsilon}_{int})) \\ (15) \quad &= \text{Cov}(\rho\tilde{\varepsilon}_{in,t-1}, \rho^2\tilde{\varepsilon}_{in,t-1}) = \frac{\rho^3\sigma_{\varepsilon_n}^2}{1 - \rho^2}, \end{aligned}$$

where F_{it} is the earnings forecast for firm i in period t . Having ρ , we can finally approximate $\tilde{\varepsilon}_{in}$ iterating [equation \(2\)](#) with lagged values of ε_{in} .¹⁸ Finally, from the approximated series $\tilde{\varepsilon}_{in}$ we can obtain our estimate for θ_{in} as

$$\hat{\theta}_{in} = F_i - Z - \rho\hat{\varepsilon}_{in}.$$

By taking the cross-sectional variance of $\hat{\theta}_{in}$ in a given year, we can readily estimate one of the components of PI, $\sigma_{\theta_n}^2$.

2. Signals About Liquidity, $\hat{\theta}_{id}$. Ideally, one would use signals about the discount a trader expects to face when selling stocks, which include the compensation to dealers for price volatility, search frictions, and inventory costs. Unfortunately, there are no forecasts of such objects; however, the discount would be proportional to the short-run volatility of the stock price in a model

17. We also use the daily exchange rates provided by I/B/E/S to standardize the currency in a market across firms, time, and variables. See [Online Appendix B.2](#) on the timing assumptions that lead to our measurements (in [Online Appendix D.3](#), we show the robustness of our timing assumptions). See [Online Appendix B.3.1](#) for details on how we assign companies to countries, [Online Appendix B.3.2](#) for sample restrictions, [Online Appendix B.3.3](#) for how we adjust data to make firms comparable, [Online Appendix B.3.4](#) for adjustments that allow us to compare firms across countries, and [Online Appendix B.3.5](#) for how we deal with mergers and acquisitions and stock splits.

18. We first estimate ρ for markets that have at least 20 companies with past year's data. For the remaining markets, we use the country average if available and, if not, the overall average. For computing $\tilde{\varepsilon}_{in}$, we determine the lag order for each company-date pair based on the availability of past data on ε_{in} , with a maximum of three lags. We initialize the $AR(1)$ process as zero.

where dealers are risk-averse profit maximizers with heterogeneous beliefs. We formalize this link in [Online Appendix C](#): a stock with high price volatility implies the dealer's risk of holding the stock is high, charging a high fee for potentially buying it high and selling it low. Based on this microfoundation, we use the prior six months of realized price volatility as an estimator for the signal about price volatility.¹⁹

$$\hat{\theta}_{id} = Vol_{t,t-5}(p_i),$$

where t is monthly. By taking the cross-sectional variance of $\hat{\theta}_{id}$, we can estimate another component of PI, $\hat{\sigma}_{\theta_d}^2$.

Having obtained the time series of signals about profitability and liquidity ($\hat{\theta}_{in}$ and $\hat{\theta}_{id}$),²⁰ we proceed to estimate their price loadings for each country-year pair, ϕ_n and ϕ_d . We first remove as much heterogeneity across firms as possible by residualizing the normalized stock prices to (i) homogenize $\bar{K}_i$ and $\bar{\theta}_{in}$ and (ii) eliminate dependencies in the distributions of θ and ε across companies (see [Online Appendix B.3.3](#) for details). Then we estimate the pricing regression [equation \(14\)](#) using the residualized stock prices for each country-year pair, $\hat{\theta}_{in}$ and $\hat{\theta}_{id}$ as our measures of θ_{in} and θ_{id} , and $\bar{\varepsilon}_i$ for firm i estimated using forecast errors obtained when constructing $\hat{\theta}_{in}$. We treat v_i as a measurement error that is orthogonal to the regressors.

Under [Assumption 1](#), the OLS estimators $\hat{\phi}_d$ and $\hat{\phi}_n$ are unbiased estimates of ϕ_d and ϕ_n for all country-year pairs. The estimates validate the model's predictions. The model suggests $\phi_n > 0$ and $\phi_d < 0$, although the estimation does not impose this restriction. The former is satisfied in all but one pair, while the latter is

19. To estimate the stock price volatility, we use the measure proposed by [Garman and Klass \(1980\)](#), which only requires the opening (O), closing (C), highest (H), and lowest (L) prices during the period. In particular, we look at the stock prices over the previous six months to compute

$$\begin{aligned} \hat{\sigma}_{it}^2 = & 0.511(H_{it} - L_{it})^2 - 0.019[(C_{it} - O_{it})(H_{it} + L_{it} - 2O_{it}) \\ & - 2(H_{it} - O_{it})(L_{it} - O_{it})] - 0.383(C_{it} - O_{it})^2 \end{aligned}$$

for each ticker i at date t . If the resulting volatility measure exceeds the stock price, we equate it to the stock price, imposing an intuitive limited-liability rule on stock ownership. We normalize this measure with the stock price. In [Online Appendix D.4](#), we provide statistics on the cross-sectional distribution of the range of volatility estimates and how they evolve for the median firm.

20. [Online Appendix Table A.4](#) show summary statistics for the United States, Japan, and the United Kingdom in 2015 as examples.

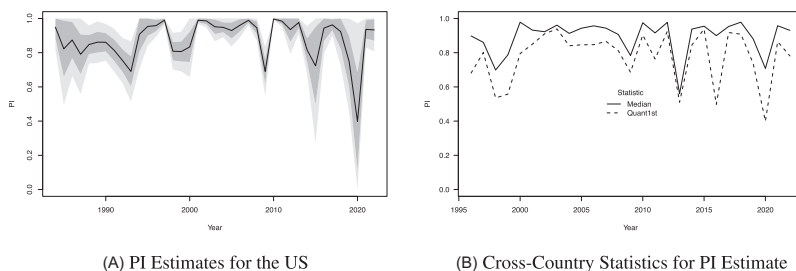


FIGURE I
The PI Estimates

In Panel B we restrict attention to countries for which PI is estimated for at least 20 years between 1994 and 2022 to get a partially balanced sample. The countries are Germany, the United Kingdom, France, Japan, Taiwan, Canada, Sweden, and the United States.

satisfied in 70% of the pairs. Furthermore, [equation \(12\)](#) predicts $\frac{|\hat{\phi}_n|}{|\hat{\phi}_d|} > 1$, which is true for 97% of the pairs.

Once we have obtained the four estimates for $\hat{\sigma}_{\theta_n}^2$, $\hat{\sigma}_{\theta_d}^2$, $\hat{\phi}_n$, and $\hat{\phi}_d$, we compute PI from [equation \(13\)](#) for 344 country-years pairs (see [Online Appendix Table A.5](#) for a list of countries and [Online Appendix Table A.6](#) for a summary of the 344 pricing regressions). Next we discuss the PI estimation results.

III.B. Price Informativeness over Time and Across Countries

As an illustration of the results, [Figure I](#), Panel A presents the PI series for the US, with one and two standard deviation bootstrap confidence bands.²¹ This series is cyclical: it has a time-series correlation of 0.51 with aggregate stock returns and 0.45 with the GDP growth rate.²² The largest declines in the 40-year sample are during the COVID-19 episode, the 2008–2009 financial crisis, and the late 1980s and early 1990s that surrounded the

21. In [Online Appendix F](#) we show the time series components behind this series: forecast error variance ($\sigma_{\varepsilon_n}^2$ and $\sigma_{\varepsilon_d}^2$ in [Online Appendix Figure A.9](#)) and signal averages ($\bar{\theta}_n$ and $\bar{\theta}_d$ in [Online Appendix Figure A.10](#)). We provide the time-series plots of estimated $\sigma_{\theta_n}^2$, $\sigma_{\theta_d}^2$, ϕ_n , and ϕ_d in [Online Appendix Figure A.8](#).

22. The cyclical patterns observed in [Figure I](#), Panel A are robust to allowing for persistence in the unlearnable part of liquidity ε_{id} , residualizing the prices with volatilities in the further past, a back-of-the-envelope bias correction for heterogeneity in θ_{in} , and controlling for ε_{in} directly to account for private information that is not captured by analyst forecasts. See [Online Appendix D.2](#) for details.

long-lived savings and loan (S&L) crisis.²³ Although we focus on the cyclical patterns of the PI measure, its levels merit precision of interpretation. PI approaching one during booms may suggest, for instance, that prices almost perfectly reveal firms' productivities. This seemingly extreme result, however, is a by-product of our stylized structure, which has only two pricing factors. Modeling additional pricing factors (beyond productivity and liquidity) naturally lowers PI levels because several other dimensions are likely conflated with a firm's productivity.

Figure I, Panel B presents moments from the yearly distribution of PI estimates across eight countries with observations between 1995 and 2021. The median and the first quartile experienced declines around the Great Recession and the COVID-19 pandemic, similar to the United States. Other major declines were observed in PI in (i) most European economies around the Taper Tantrum and the European sovereign debt crisis in 2013 and (ii) the United Kingdom and France around the Brexit decision in 2016.²⁴

An advantage of our structural estimation of PI is that we can identify the drivers behind these large PI declines by following the four components individually. A clean decomposition can be obtained by considering the log of a monotonic transformation of PI from equation (13):²⁵

$$(16) \quad \ln\left(\frac{PI}{1-PI}\right) = \ln(\phi_n^2) + \ln(\sigma_n^2) - \ln(\phi_d^2) - \ln(\sigma_d^2).$$

Figure II, Panel A presents this PI transformation for the United States (in the solid black line) together with two components: price loadings and variances. The variances (the dotted line) are important for the level of PI but play a relatively small role in its

23. Even though there was no systemic disruption of financial institutions in 2014–2015, there was a deterioration in other markets that also provide liquidity, such as Treasuries and corporate bonds. First, the yield on Treasuries declined steadily after 2009, reaching lows in 2014, with a 35 basis-point collapse on October 15, 2014, known as the “flash crash.” Second, dealer inventories of corporate bonds also reached very low levels, making it difficult to execute large trades in 2014.

24. Average PI levels do not differ much across countries, even in a larger set of countries (See [Online Appendix Table A.6](#)), with no clear relationship between the country's (or its stock market's) development and the estimated level of PI. This pattern is not at odds with our theory: PI is not about the absolute number of stock market participants or the total amount spent on information acquisition; it is about the relative attention paid to firms' fundamentals versus stock liquidity.

25. We thank Liyan Yang for suggesting this intuitive decomposition.

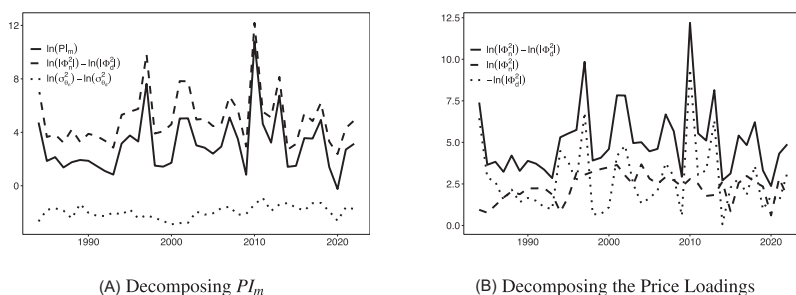


FIGURE II
Contribution of the Components of PI

fluctuations relative to price loadings (the dashed line). Figure II, Panel B further decomposes price loadings. While fluctuations in the loading of earnings (dashed line) contribute significantly, the lion's share of PI fluctuations comes from changes in the loading of liquidity (dotted line). An alternative to understand the role of each component on PI's fluctuations is: "How much would the time series variance of the PI measure decline if the component x is kept fixed at its median value?" The variance would decline by 5%, 20%, and 72% with $\sigma_{\theta_d}^2$, ϕ_n , and ϕ_d at their median values, respectively. If $\sigma_{\theta_n}^2$ instead was kept at its median value, the PI variance would increase by 24%. This is just a statistical decomposition as ϕ_n and ϕ_d are themselves functions of $\sigma_{\theta_n}^2$ and $\sigma_{\theta_d}^2$. In the next section, we make it structural.

We now test whether the cyclicity of PI (information cycles) relates to usual measures of business cycles and liquidity cycles. We run the following regression:

$$(17) \quad PI_m = \beta_0 + \beta_1 Cycle_m + CountryFE_m + \epsilon_m,$$

where m denotes a market, that is, a country-year pair. To measure business cycles ($Cycle_m$), we use two proxies: aggregate earnings and GDP growth.²⁶ To focus on cyclicity, we add country fixed effects to account for PI's country-specific factors. Table I, columns (1)–(4) present the results. There is a positive correlation between both measures of economic activity and the estimated PI series.

26. We use growth rate of GDP measures from the World Bank. Taiwan's GDP growth measure is provided by its National Statistics Bureau. We apply linear detrending. See Online Appendix Table A.7 for summary statistics of economic conditions in this panel of countries.

TABLE I
THE CYCLICALITY OF THE PRICE INFORMATIVENESS MEASURES

	PI					
	(1)	(2)	(3)	(4)	(5)	(6)
GDP growth rate	0.47 (0.48)		2.55*** (0.38)			
Avg. earnings		3.58*** (1.27)		7.71*** (0.97)		
Banking stock perf.					0.28 (0.47)	0.66** (0.30)
Range	'84-'22	'84-'22	'84-'22	'84-'22	'84-'22	'84-'22
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Weights	No	No	Yes	Yes	No	Yes
Observations	344	344	344	344	319	319

Notes. In columns (3), (4), and (6), each country-year observation is weighted with the number of stocks used in the estimation of the PI measure. Standard errors are in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$.

The correlation becomes larger and more precisely estimated once each country-year observation is weighted by the number of stocks used to estimate PI in that country-year pair (columns (3) and (4)). In other words, stock markets reveal more information about firms when the economy is stronger.

To measure liquidity cycles, we use the ratio of the median normalized stock price of publicly listed banks to that of publicly listed nonfinancial firms, which we call banking stock performance. This measure captures how well the banking sector is performing relative to the rest of the economy, which likely affects the availability of credit (a proxy for $1 - \gamma$ in the model). Table I, columns (5) and (6) show the results. Consistent with our model, stock prices become more informative about firms’ fundamentals when stock markets have a lesser role in liquidity provision. In other words, stock markets reveal more information about firms when banks are healthier.

IV. MODEL CALIBRATION

We have measured PI for different countries and documented its cyclical nature. Here, we examine the effect of its fluctuations on real variables, arising from agents using $E[\theta_{in}|p_i]$ rather than θ_{in} to allocate capital. The expression for the output loss due to misallocation is a complicated object, but is monotonic in the Bayesian risk associated with using $E[\theta_{in}|p_i]$ as an estimator for

θ_{in} . To compute this loss, we proceed to calibrate the full model to the United States.²⁷

Our model focused on prices as the single source of information for capital allocation, but in reality, decision makers rely on additional sources, which likely mitigates the importance of PI on real outcomes. To discipline the ultimate role of PI, we extend our model. Instead of assuming the mutual fund makes decisions solely based on observing the theoretical prices p_i we assume it observes the same prices as the econometrician, p_i^s , an independent signal about the learnable component of firm i 's fundamentals $\theta_{in}^s \sim N(\theta_{in}, \sigma_{\theta_{in}}^2)$ and an independent signal about the non-learnable component to traders $\varepsilon_{in}^s \sim N(\varepsilon_{in}, \sigma_{\varepsilon_{in}}^2)$. In short, the mutual fund makes decisions based on $E[z_{in}|p_i^s, \theta_{in}^s, \varepsilon_{in}^s]$ instead of $E[z_{in}|p_i]$.

These additions acknowledge that decision makers recognize that prices contain additional noise and that they may possess information that is independent of and distinct from that of traders. In other words, while our empirical estimate captures how much prices inform decision makers about what informed traders know, our calibration captures the extent to which that information is new to decision makers and is then incorporated into allocations: we capture revelatory PI.

IV.A. Calibration Strategy

We already estimated the time series of the means ($\bar{\theta}_n, \bar{\theta}_d$) and the variances ($\sigma_{\theta_n}^2, \sigma_{\theta_d}^2, \sigma_{\varepsilon_n}^2$, and $\sigma_{\varepsilon_d}^2$). Here we finalize the model's parameterization. We start by externally calibrating $\{\bar{k}, \xi, \eta, \alpha, \delta, \rho, r^F\}$. Given the lack of feedback from capital allocation to stock markets in our model, we can conduct the internal calibrations sequentially. In a first stage, we internally calibrate parameters involved in the operation of stock markets, the latent series of γ_t , and the information cost parameters $\{v_d, v_n, \psi_n, \psi_d\}$. In the second stage, we internally calibrate the parameters related to capital allocation $\{\beta, \sigma_{p_i^s}, \sigma_{\theta_{in}^s}, \sigma_{\varepsilon_{in}^s}\}$ to discipline the extent to which prices affect the real sector.

27. As a benchmark, in [Online Appendix E](#), we show the analytical link between PI and misallocation losses under a quadratic loss function, which is not the loss function that arises in our setting.

1. *Externally Calibrated Parameters.* For households, we set the intertemporal elasticity of substitution $\frac{1}{\eta}$ equal to 0.5 consistent with Gruber (2013). For traders, following Farboodi and Veldkamp (2020), we set the absolute risk aversion parameter a equal to 0.05. We set the depreciation rate δ equal to 0.06 (see Bai et al. 2022), the risk-free interest rate available to the traders r^F equal to 0.02, and the autocorrelation of the unlearnable part of profitability, ρ , to 0.²⁸ Last, because we directly estimated productivity from earnings data, we have some flexibility in setting the scale of the economy. So we normalize the adjustment cost $\frac{\bar{k}}{\varepsilon}$ to achieve an average capital level of one.

2. Internally Calibrated Parameters of Stock Markets.

i. *Series of Liquidity Needs, γ_t .* Equation (12) links γ_t to the ratio of pricing coefficients $\frac{|\phi_{in}|}{|\phi_{id}|}$, the ratio of residual payoff variances $\frac{\sigma_{\varepsilon_{id}}^2}{\text{Var}(z_{in} + p'_{in})}$ and the ratio of informed traders $\frac{\lambda_n}{\lambda_d}$. While the ratio of pricing coefficients and $\sigma_{\varepsilon_{id}}^2$ were estimated in Section III to compute PI measures, estimating $\text{Var}(z_{in} + p'_{in})$ involves capturing the uncertainty about future prices, which we approximate by estimating a first-order autoregressive process for the estimated pricing coefficient vector $\hat{\Phi}$:

$$(18) \quad \hat{\Phi}_t = B_0 + B_1 \hat{\Phi}_{t-1} + W_t,$$

where $\hat{\Phi}_t = [\hat{\phi}_0 \ \hat{\phi}_\varepsilon \ \hat{\phi}_n \ \hat{\phi}_d]$, B_1 is a 4×4 diagonal matrix that controls the persistence, B_0 is a 4×1 vector that stores the constant terms, and W_t is a 4×1 error term where $W_t \sim MVN(0, Q)$ and Q is the associated variance-covariance matrix. Once estimated, equation (18) provides a simple formula for $\text{Var}(z_{in} + p'_{in})$.

After recovering $\text{Var}(z_{in} + p'_{in})$, we use Lemma 1 to obtain $\frac{\lambda_n}{\lambda_d}$. In an interior equilibrium, the information acquisition conditions boil down to the following equations for night and day traders:

28. Even though we computed PI estimating $\rho > 0$, here we assume $\rho = 0$. Otherwise, the unlearnable component of productivity ε_i never reaches an ergodic distribution under aggregate shocks, and its distribution becomes an additional state variable. We will use the calibration of θ_{in}^s to account for the missing ρ quantitatively. Still, for aggregate outcomes, the identity of the firms with high productivity in each period is irrelevant, turning ρ irrelevant in assessing the allocative implications of PI.

$$(19) \quad e^{ac_n(\lambda_n)} \text{Var}(\varepsilon_{in} + p'_i) = \text{Var}(\varepsilon_{in} + p'_i) + \frac{\sigma_{\theta_n}^2}{1 + \frac{\sigma_{\theta_n}^2}{\sigma_{\phi_n}^2} \left(\frac{\phi_n}{\phi_d}\right)^2},$$

$$(20) \quad e^{ac_d(\lambda_d)} (\sigma_{\varepsilon_d}^2 + \text{Var}(\varepsilon_{in} + p'_i)) = (\sigma_{\varepsilon_d}^2 + \text{Var}(\varepsilon_{in} + p'_i)) + \frac{\sigma_{\theta_n}^2}{1 + \frac{\sigma_{\theta_n}^2}{\sigma_{\phi_d}^2} \left(\frac{\phi_n}{\phi_d}\right)^2} + \frac{\sigma_{\theta_d}^2}{1 + \frac{\sigma_{\theta_d}^2}{\sigma_{\phi_n}^2} \left(\frac{\phi_d}{\phi_n}\right)^2}.$$

These two equations pin down λ_n and λ_d for given cost functions $c_n(\cdot)$ and $c_d(\cdot)$.

ii. Information Cost Functions. We parameterize the information cost function as follows:

$$(21) \quad c_j(\lambda_j) = v_j \left(\frac{1}{1 - \lambda_j} \right)^{\psi_j} - v_j$$

for $j \in \{d, n\}$, which satisfies $c_j(0) = 0$ and $\lim_{x \rightarrow 1} c_j(x) = \infty$, and guarantees an equilibrium with $\lambda_j \in [0, 1)$.

For each set of parameters, $\{v_d, v_n, \psi_d, \psi_n\}$ there is a mapping between the loadings (ϕ_n and ϕ_d) and the fraction of informed traders (λ_n and λ_d) through [equations \(19\) and \(20\)](#). We choose these four parameters to match four moments implied by the estimated pricing function: (i) the level of information acquisition by night and day traders, (ii) the increase in information acquisition in recessions, and (iii) the level of PI. The level of the cost of acquiring information depends on v_d and v_n , whereas its curvature is determined by ψ_d and ψ_n . The average levels of λ_n and λ_d are hence directly informative about v_d and v_n . The change in information acquisition activities, on the other hand, reveals the curvatures. Finally, the average level of PI informs how day traders respond to information acquisition by night traders: a low ψ_d (ψ_n) makes it easier for a day (night) trader to scale his information acquisition, tilting the prices to reflect more of θ_d (θ_n).

iii. Discretizing the Aggregate States. To estimate the aggregate state ($s_t = \{Z_t, \gamma_t\}$) dynamics, we remove a linear trend from the productivity series Z_t (measured by the log of earnings per share) and the liquidity needs series γ_t (recovered as described above). We estimate a vector autoregression (VAR) and discretize it with a four-state Markov chain (two levels for each state) following [Gospodinov and Lkhagvasuren \(2014\)](#). We assign each year as

a high or low Z given the estimated Z grid and compute the average levels of $\sigma_{\theta_n}^2$, $\sigma_{\theta_d}^2$, $\sigma_{\varepsilon_n}^2$, and $\sigma_{\varepsilon_d}^2$ in those years. These averages correspond to variances in normal times and recessions.

iv. *Estimation Algorithm for the Stock Market.* To be more precise about the calibration procedure, we start with a guess $\{\nu_d^0, \nu_n^0, \psi_d^0, \psi_n^0\}$ and take the next steps:

- i. Start iteration k with $\{\nu_d^k, \nu_n^k, \psi_d^k, \psi_n^k\}$. Use equations (19) and (20) with the estimated series for pricing coefficients and parameters to infer the $\lambda_{d,data}^k$ and $\lambda_{n,data}^k$ series.
- ii. Invert equation (12) to infer the γ series.
- iii. Follow the discretization procedure discussed above to estimate a VAR for $\{Z, \gamma\}$ and discretize it into a Markov chain. Compute the values for $\sigma_{\theta_n}^2$, $\sigma_{\theta_d}^2$, $\sigma_{\varepsilon_n}^2$, and $\sigma_{\varepsilon_d}^2$ associated with each Z level. Call these values M_{data}^k .
- iv. Compute the market equilibrium and the model's counterparts of the moment values above. Call these values M_{model}^k .
- v. Compare M_{model}^k with M_{data}^k . If the discrepancy is below the threshold, stop. If not, go back to step (i) with new $\{\nu_d^{k+1}, \nu_n^{k+1}, \psi_d^{k+1}, \psi_n^{k+1}\}$.

3. *Internally Calibrated Parameters of the Real Sector.* Now, we internally calibrate parameters involved in capital accumulation and allocation: β , $\sigma_{p_i^s}^2$, $\sigma_{\varepsilon_{in}^s}^2$, and $\sigma_{\theta_{in}^s}^2$. We estimate the discount factor to generate a 2% risk-free interest rate.²⁹ The last three parameters are not in the benchmark model, but are quantitatively relevant in disciplining how much decision makers react to prices and their information content. We discipline $\sigma_{p_i^s}^2$ directly with the R-squared of the price regression from equation (14), which indicates what portion of the variation in observed prices can be explained by θ_{in} and θ_{id} , and hence the residual contained in observed prices p_i^s .

We jointly discipline the quality of mutual fund own signals, $\sigma_{\varepsilon_{in}^s}^2$ and $\sigma_{\theta_{in}^s}^2$ using the cross-sectional price-investment correlation ($cor(p_i^s, k_i)$), and the time-series correlation between PI and invest-

29. This interest rate is not the return households receive from capital (r) because the realized return accrues all the surplus to the households. Instead, we evaluate the counterfactual interest rate in the model that would clear markets in a competitive setting.

ment dispersion ($\text{cor}(PI, \text{std}(k_i))$). First, $\text{cor}(p_i^s, k_i)$ increases with both variances: the mutual fund pays more attention to prices when making a decision when their own signals are less informative. Second, $\text{cor}(PI, \text{std}(k_i))$ also increases with both variances: when the mutual fund pays more attention to prices, PI fluctuations have a smaller effect on capital allocation measured by the cross-sectional dispersion of investment. Even though both effects go in the same direction, the elasticity of the latter with respect to $\sigma_{\theta_{in}^s}^2$ is much larger because the mutual fund's own signal about θ_n is correlated with prices, providing a single crossing condition for identification.³⁰ Intuitively, both signals put limits on the extent of learning from the prices. As the precision of θ_{in}^s increases, prices become a sideshow, and fluctuations in PI are irrelevant for capital allocation. As the precision of ε_{in}^s increases, the investment decision assigns a smaller weight to stock prices, since part of the investment choices is based on the realization of ε_{in}^s .

Although our benchmark suggests a tight link between prices and adjustable capital, this relation has been empirically challenged. [David, Hopenhayn, and Venkateswaran \(2016\)](#), for instance, documented a weak cross-sectional correlation between capital expenditures (CAPEX) and stock prices, arguing that learning from prices must be limited. As shown in [Online Appendix Figure A.11](#), our data suggests a similar weak correlation (in more recent years, this correlation is even negative).³¹ It has been noted that CAPEX primarily captures expenses on fixed long-term assets (such as buildings and structures) that tend to be sluggish and multifaceted. This concern is further supported by the observed weak correlation between CAPEX and other variables that are presumably tightly connected to a firm's investment ability and decisions, such as previous earnings, earnings forecasts, and future earnings.

When alternative measures of investment, or proxies for the capacity to attract investment funds, are used, the relationship with prices becomes intermediate: not absent, as with CAPEX, nor

30. Our identification relies on $\sigma_{\varepsilon_{in}^s}^2$ and $\sigma_{\theta_{in}^s}^2$ being constant over time. It is impossible to rule out a source of information that independently provides the same information and fluctuates similarly to stock prices, yet doesn't react to counterfactual policies that affect stock markets.

31. In the exercises described in this section, we transform all variables to per-unit-of-asset values to allow comparability across firms, similar to our empirical exercise in [Section III](#). See [Online Appendix B.3.3](#) for details.

perfect, as in our benchmark. To be more precise, we consider three alternative measures of investment: (i) asset changes, computed as the year-to-year change in total assets,³² (ii) working capital, and (iii) the number of employees. We also consider two indicators of how easily a firm can access funds: corporate bond yields and corporate bond ratings. [Online Appendix Figure A.12](#) shows that for the United States in 2015, while CAPEX has a weak negative correlation with stock prices, the other measures tell a different story. Firms with higher stock prices tend to have larger changes in assets, have higher working capital and employee levels, pay lower rates on their bonds, and have better-rated bonds. [Online Appendix Tables A.8](#) and [A.9](#) also support a tight connection: prices predict higher measures of productive capacity (including a higher CAPEX) once previous earnings, country, and time fixed effects are controlled, with and without firm fixed effects.

We consider working capital to be closest to the perfectly adjustable capital in our model. Working capital does not necessarily capture all productive capital, but it is a reactive proxy for short-term funds and operational flexibility that enables firms to exploit opportunities. In what follows, we use working capital as our measure of k_i .

IV.B. Calibration Results

[Figure III](#), Panel A shows the estimated liquidity series for the United States, suggesting elevated reliance on stock markets for liquidity purposes (high γ) around the Great Recession and the COVID-19 pandemic and a persistent period of high reliance surrounding the period of S&L crisis from 1985 to 1995.

[Figure III](#), Panel B shows the calibrated shape of information cost functions for day and night traders. The cost of learning for day traders is much higher and steeper than that for night traders. This result is disciplined by two observations: (i) the uncertainty conditional on learning about stock liquidity ($\sigma_{\varepsilon_d}^2$) is much higher than the corresponding uncertainty in fundamental payoffs ($\sigma_{\varepsilon_n}^2$), and (ii) the price uncertainty ($\text{Var}(\varepsilon_{in} + p'_i)$) is much larger than the dispersion of signals ($\sigma_{\theta_n}^2$ and $\sigma_{\theta_d}^2$). From [equations \(19\)](#) and

32. Unlike CAPEX, which measures cash outflows for long-term assets, asset change measures the net change in a firm's productive capacity while taking into account current assets and mergers and acquisitions. Indeed, [Cooper, Gulen, and Schill \(2008\)](#) argues that asset growth is a better predictor of future stock returns.

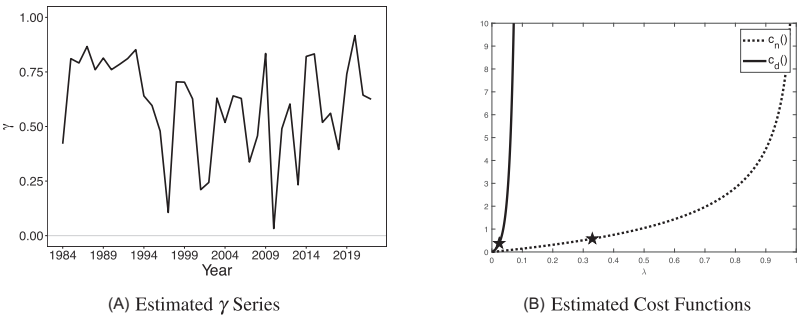


FIGURE III
 γ Series and Information Acquisition Costs

Panel A presents the estimated γ series. Panel B shows estimated information acquisition cost functions. Solid and dashed lines represent $c_d(\cdot)$ and $c_n(\cdot)$, respectively. Stars indicate the median λ and associated costs in the stochastic steady state.

TABLE II
ESTIMATED AGGREGATE STATE LEVELS

s	Z	γ	σ_{θ_n}	σ_{θ_d}	σ_{ε_n}	σ_{ε_d}	Years
1	0.049	0.35	0.059	0.17	0.036	0.14	'95–'97, '01, '02, '13, '16
2	0.049	0.79	0.059	0.17	0.036	0.14	'89, '90, '91–'94, '98–'00, '03, '09, '15, '17, '19–'21
3	0.061	0.35	0.059	0.14	0.038	0.12	'84, '04, '07, '08, '10, '11, '18
4	0.061	0.79	0.059	0.14	0.038	0.12	'85–'88, '05, '06, '12, '14, '22

Notes. The last column shows the years with Z and γ estimates closest to the values associated with each aggregate state. The values of $\sigma_{\theta_n}^2$, $\sigma_{\theta_d}^2$, $\sigma_{\varepsilon_n}^2$, and $\sigma_{\varepsilon_d}^2$ are estimated by taking the averages over the years associated with the Z values.

(20), this is consistent with night traders' lower information costs. If information costs functions were the same, day traders would obtain more information than night traders, given the additional uncertainty they face, with a ratio given by equation (12) counterfactually low.

Table II summarizes the four aggregate states resulting from discretizing the (Z, γ) series and the years our method assigns to each aggregate state. We also show the signal and forecast error standard deviations for each state. Z fluctuates between 0.049 and 0.061 (consistent with median earnings per share fluctuations) while γ fluctuates between 0.35 and 0.79. We interpret the values of γ as indicating that roughly two-thirds of traders focus exclusively on stocks' long-run earnings when banks operate well,

whereas most focus on short-run price fluctuations when funding liquidity dries up.

Table III summarizes all calibrated parameters, externally and internally. We set PI corresponding to the aggregate state $\{\bar{Z}, \gamma\}$, i.e., $\underline{PI}$, as a benchmark.

Finally, in Table IV we validate our calibration and show it is consistent with PI fluctuations, which we do not explicitly target. Ceteris paribus, PI declines when the economy transitions to a state with higher market liquidity needs and increases when the economy transitions to a state with lower aggregate productivity. If both productivity declines and banking gets into distress, the first effect prevails, and PI declines. These changes are consistent with the PI's cyclical properties when productivity and liquidity are correlated.

V. QUANTITATIVE RELEVANCE OF STOCK PI

This section uses the calibrated model to assess the quantitative relevance of stock PI on economic activity. The first exercise estimates an impulse response function for a recessionary shock, with and without financial market distress. The goal is to assess whether financial distress amplifies or dampens the effect of a recessionary shock by altering the informativeness of stock prices. The second exercise compares the aggregate effects of recessions with financial distress in alternative economies, one with lower information costs and another in which traders receive information exogenously. Whereas the first economy is useful for evaluating transparency policies, the second quantifies the role of endogenous information acquisition in the economic response to changing trader composition. Last, in a third exercise, we quantify the importance of alternative sources of information through economies with various combinations of prices, signals on the learnable component of productivity (θ_n^s), and signals on the unlearnable component of productivity (ε_n^s).

V.A. Allocation Effects of Productivity and Liquidity Shocks

We simulate our economy for high Z and low γ (state 3 in Table II). We introduce a one-period recession: a decline in Z (a transition to state 1 in Table II). This single shock captures standard productivity recessions, without clear liquidity problems. We compare this scenario to a one-period recession with financial dis-

TABLE III
EXTERNALLY AND INTERNALLY CALIBRATED PARAMETERS

Parameter	Value	Moment	Parameter	Value	Moment	Model	Target
$\bar{z}$	174.6	Normalized	v_n	4.58	λ_d	0.06	0.02
$\bar{\xi}$	2	External	v_d	0.11	λ_n	0.26	0.33
η	0.05	External	ψ_n	0.3	$\overline{PI}$	0.87	0.87
α	0.06	External	ψ_d	60	$\Delta\lambda$	0.23	0.22
δ	0.02	External	β	0.98	Real interest rate	0.02	0.02
r^F			$\sigma_{PI_t^s}$ for γ	4.9	R -squared for γ	0.44	0.44
			$\sigma_{PI_t^s}$ for $\bar{\gamma}$	2.5	R -squared for $\bar{\gamma}$	0.44	0.44
			$\sigma_{\varepsilon_{jn}^s}$	0.02	$cor(p_i, k_i)$	0.39	0.37
			$\sigma_{\psi_{jn}^s}$	0.11	$cor(PI, std(k_i))$	0.22	0.22

Notes. λ_d and λ_n are computed as the average values in the ergodic distribution of aggregate states. The data counterpart is computed as the median value in the inferred series. $\overline{PI}$ is the PI level averaged over low Z states in the ergodic distribution. Its data counterpart is computed as the average PI level in designated low Z years. The $\Delta\lambda$ is computed as the percentage change in aggregate (day and night) fraction of informed traders in an IRF exercise where the economy moves from a long sequence of high Z , low γ states to a state of low Z , high γ . Its data counterpart is the percentage difference in employment in the NAICS 52394 (portfolio management and investment advice) industry during recessions and expansions (CES Survey, BLS).

TABLE IV
UNTARGETED PI MOMENTS

Moment	Model	Untargeted
$\Delta PI_{\bar{z}\gamma \rightarrow z\gamma}$	0.02	0.01
$\Delta PI_{\bar{z}\gamma \rightarrow \bar{z}\bar{\gamma}}$	-0.16	-0.10
$\Delta PI_{\bar{z}\gamma \rightarrow z\bar{\gamma}}$	-0.10	-0.17

Notes. The ΔPI terms are computed as percentage changes in PI measures in an IRF exercise where the economy moves from a long sequence of high Z , low γ states. Their data counterpart is computed as the percentage differences in average PI levels across years with different aggregate state designations.

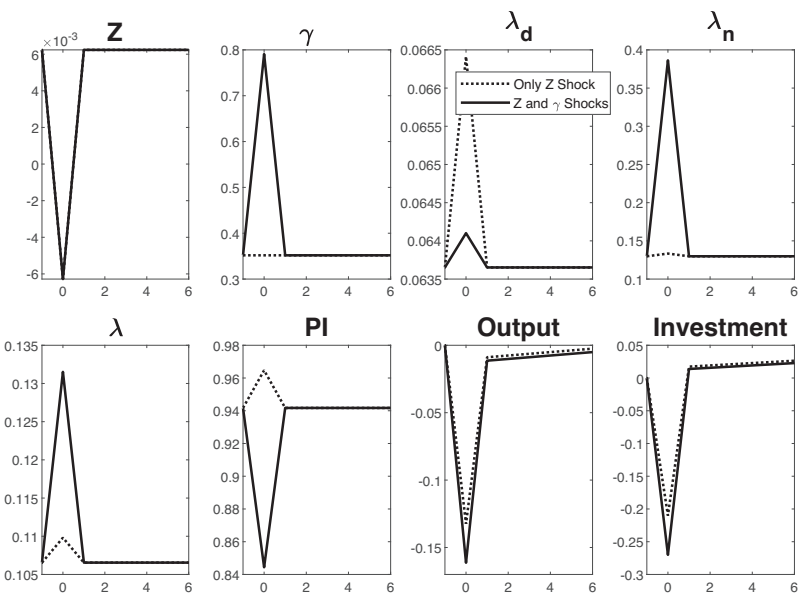


FIGURE IV
Impulse Response Functions

The first two panels provide the shocks that hit the economy under the benchmark and the counterfactual recession scenarios. λ denotes the aggregate fraction of traders that acquire information. Output and investment values represent the percentage changes from the pre-shock values.

stress: a decline in Z and an increase in γ (a transition to state 2 in Table II). This dual shock captures the major downturns of the U.S. economy during our estimation period: the Great Recession in 2009 and the COVID-19 recession in 2020. Figure IV presents the

implications of these two different aggregate shocks: the dotted line is the first case, and the solid line is the second.³³

In the case of a single shock to productivity (dotted lines), the increase in uncertainty σ_{θ_d} that accompanies a recession induces day traders to produce more information, but the increase in σ_{ε_d} makes them trade less aggressively on their information. This combination generates less “noise” in markets and makes stock prices more informative about fundamentals. The increase in PI improves capital allocation, partially compensating for the negative productivity shock, which, alone, without a change in PI, would have generated an output loss 4.4% larger.

What if the recession is accompanied by financial distress? With dual shocks to productivity and liquidity needs (solid lines), the participation of more day traders obscures firms’ fundamentals. As a response to more uncertain fundamentals, the benefits of acquiring information also increase for night traders. Despite overall increased information acquisition (higher $\lambda = \gamma\lambda_d + (1 - \gamma)\lambda_n$), PI ultimately declines, inducing a decline in output and gross investment that is 22% larger than without financial distress.³⁴

1. Testable Implications. Our calibrated model generates a relation between shocks, PI, and allocations that we can empirically test across countries. First, our calibration shows that PI rises with productivity shocks and falls with financial distress and reacts more to standard financial distress than to standard productivity shocks. To test these predictions, we estimate:

$$(22) \quad PI_{ct} = \beta_0 + \beta_1 Z_{ct} + \beta_2 y_{ct} + F_c + F_t + \epsilon_{ct},$$

where PI_{ct} is our PI measure for country c in year t , Z_{ct} represents the business cycle measured by GDP growth, and y_{ct} represents the funding liquidity cycle. For y_{ct} , we use the banking stock performance variable, which we presented in Table I, and several other proxies, such as binary measures of banking panics and banking equity crises (constructed by Baron, Verner, and

33. The model is calibrated to the U.S. publicly listed sector, therefore, the fluctuations are larger in magnitude than the output fluctuations implied by national accounts.

34. In Online Appendix Figure A.13, we compare a banking crisis with and without a drop in Z . Whereas in both cases PI declines, this is more pronounced when there is no recession, leading to a pure misallocation decline in output (despite no reduction in Z) that is 11% of a standard recession.

Xiong 2021), and continuous measures, such as capital-asset ratios of the banking sector, loan spreads of the banks (lending rate minus Treasury bill rate), and the ratio of nonperforming loans to total gross loans. In low-liquidity environments, such as the Great Recession, the capital-asset ratio is expected to be low, while the loan spreads and nonperforming ratios are expected to be high. Finally, F_c and F_t are country and year fixed effects, respectively.

Table V presents the results. Consistent with our calibration, lower liquidity from the banking and financial sectors is associated with lower PI across all specifications. A one standard deviation increase in the capital-to-asset ratio of the banking system, for example, is associated with a 0.2 increase (0.83 standard deviations) in PI. That is, when banks are better armed to provide credit, stock prices become more informative. However, once conditioned on the level of economy-wide liquidity, lower aggregate productivity is associated with higher PI, but this association is not statistically significant.³⁵

Second, misallocation from a drop in PI takes a particular form in our model: lower PI induces firms' fundamentals to look more alike, reducing investment dispersion, all else equal. We test this implication by comparing the dispersion of capital expenditures in a particular country-year pair against the level of PI in Table VI. Regardless of how country-year pairs are weighted, a higher PI is associated with greater dispersion in capital expenditures, even after controlling for the business cycle.

Our result of a recession without financial distress is reminiscent of cleansing recessions, but for different reasons. In that literature, recessions reduce the cost of reallocating resources to more productive activities. In our case, recessions increase stock market informativeness and improve resource allocation. Our work, however, highlights that cleansing recessions can turn into sullying recessions when accompanied by heightened liquidity concerns. Recessions accompanied by weakness in the banking sector are periods of lower PI and worse resource allocation. In prior literature, tighter borrowing constraints, time-varying risk premia, countercyclical adverse selection in the market for used capital, and managers' incentives to hide reallocation needs during

35. Most estimates are robust to using aggregate earnings as the measure of economic activity (see Online Appendix Table A.10) or to weighting observations with the number of stocks used in the PI estimation (see Online Appendix Table A.11).

TABLE V
PRICE INFORMATIVENESS AND ECONOMIC CONDITIONS

	PI					
	Banking stock perf. (1)	Bank capital to assets (2)	Bank loan spreads (–) (3)	Nonperforming loans (–) (4)	Banking panic (–) (5)	Banking equity crisis (–) (6)
GDP growth	–0.99 (0.85)	–0.64 (1.20)	–0.36 (1.39)	–1.29 (1.47)	–1.32 (0.99)	–1.35 (0.99)
Liq. measure	0.49 (0.56)	0.09** (0.04)	0.02*** (0.01)	0.03*** (0.01)	0.13* (0.07)	0.01 (0.07)
Range	'84–'22	'05–'22	'84–'22	'05–'22	'84–'16	'84–'16
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	319	180	185	185	244	244

Notes. In each regression, the dependent variable is the PI. Column labels refer to the liquidity measure used in each regression. Both country and year fixed effects are included. The standard errors are clustered at the country level. * $p < .1$; ** $p < .05$; *** $p < .01$.

TABLE VI
PI AND CAPITAL ALLOCATION

	Dispersion of Investment			
	(1)	(2)	(3)	(4)
PI	0.004** (0.002)	0.004** (0.002)	0.005** (0.002)	0.005** (0.002)
GDP growth		0.036** (0.018)		0.060* (0.031)
Range	'84-'22	'84-'22	'84-'22	'84-'22
FE	Yes	Yes	Yes	Yes
Weights	No	No	Yes	Yes
Observations	344	344	344	344

Notes: In each regression, the dependent variable is the standard deviation of normalized capital expenditures. Both country and year fixed effects are included. In columns (3) and (4), each country-year observation is weighted with the number of stocks used to estimate the PI measure. The standard errors are clustered at the country level for the unweighted regressions, and are presented in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$.

recessions have been proposed as potential mechanisms for countercyclical misallocation.³⁶ Ours is a novel mechanism that can generate higher or lower misallocation depending on whether a recession coincides with distress in the financial sector.

Third, according to our model, information acquisition increases during downturns, whether or not funding liquidity distress is present. This is consistent with the findings of [Loh and Stulz \(2018\)](#) and [Jiang, Habib, and Gong \(2015\)](#). The former documents that financial analysts produce longer, more frequent reports, and that these reports have a larger price effect in bad times. The latter documents that management earnings forecasts become more frequent during recessions. Our model rationalizes these patterns as endogenous responses to less informative stock prices: there may be less information available for decision making despite more information being generated overall.

V.B. Alternative Information Structures

In the previous exercise, we analyzed the quantitative response of PI to productivity and liquidity shocks, as well as their

36. See [Ordoñez \(2013\)](#), [Khan and Thomas \(2013\)](#), [Fajgelbaum, Schaal, and Taschereau-Dumouchel \(2017\)](#), and [Straub and Ulbricht \(2024\)](#) for tighter financial constraints; [David, Schmid, and Zeke \(2022\)](#) for time-varying risk premia; [Fuchs, Green, and Papanikolaou \(2016\)](#) for adverse selection; and [Eisfeldt and Rampini \(2008\)](#) for managerial incentives.

TABLE VII
COUNTERFACTUAL ESTIMATES

Moments	Baseline	Fixed λ	low ν_n	low ν_d
Panel A: Impulse response functions				
$\Delta \lambda_{\bar{z}\gamma \rightarrow \bar{z}\bar{\gamma}}$	0.234	0.000	0.195	0.227
$\Delta PI_{\bar{z}\gamma \rightarrow \bar{z}\bar{\gamma}}$	-0.103	-0.624	-0.067	-0.121
$\Delta Y_{\bar{z}\gamma \rightarrow \bar{z}\bar{\gamma}}$	-0.161	-0.230	-0.153	-0.164
$\Delta Inv_{\bar{z}\gamma \rightarrow \bar{z}\bar{\gamma}}$	-0.270	0.362	-0.241	-0.271
Panel B: Stochastic steady-state averages				
Y	0.110	0.099	0.113	0.109
C	0.050	0.044	0.051	0.050
Inv	0.060	0.055	0.061	0.060
PI	0.865	0.607	0.913	0.847
λ_n	0.254	0.130	0.328	0.272
λ_d	0.064	0.082	0.065	0.074

Notes. In Panel A, each number denotes the percentage change at the moment when the economy moves from a long sequence of $\bar{z}_\gamma$ states to a $\bar{z}\bar{\gamma}$ state. For the fixed γ scenario, this is equivalent to moving to $\bar{z}\bar{\gamma}$.

aggregate consequences. Here we compare these reactions with those of alternative economies. The first column of Table VII, Panel A shows the changes in total information acquisition (λ), PI, output, and investment when the economy suffers the dual shock on productivity and liquidity needs. These changes are expressed as percentages relative to the benchmark aggregate state given by high Z and low γ . In this baseline economy, for instance, a dual shock that reduces Z and increases γ reduces PI by 10% and output by 16%. These numbers replicate the magnitudes of the changes depicted by the solid lines of Figure IV.

One alternative economy assumes that a fraction of traders receive signals exogenously, captured in the second column of Table VII, Panel A, where λ always equals the amount in the boom periods of the benchmark economy. This economy would suffer a 62% reduction in PI (six times larger than the benchmark with endogenous information), leading to a severe misallocation that reduces output by 23% and investment by 36%. This result highlights the quantitative relevance of endogenous information acquisition. In the economy, agents respond to lower information content in stock markets by acquiring more information, partially offsetting the reduction. Information acquisition provides a stabilizing force against recessions with financial distress.

Another alternative economy we consider is one with lower information costs. Because two types of traders acquire different

information in our setting, we consider two situations, shown in the last two columns of [Table VII](#), Panel A. If night traders can acquire information at half the cost (third column), they would trade more aggressively; PI would decline only 6.7% instead of 10%, leading output to decline 15.3% instead of 16.1%. In contrast, if day traders could acquire information at half the cost (last column), they would also trade more aggressively, PI would decline by 12.1% rather than 10%, and output would decline by 16.4%. Note that in our setting, more information acquisition does not mean better information; it just means more traders of a particular type become informed and trade more aggressively on that information, thereby affecting which signal is “more represented” in prices. Therefore, additional information made available to traders can create negative externalities by suppressing other information, as in [Banerjee, Davis, and Gondhi \(2018\)](#).

An economy with lower information costs for night traders, v_n , could represent, for instance, improvements in the Securities and Exchange Commission disclosure regulations, with more information requirements to be filed with each prospectus, more frequent filings, or firm-specific statistics publications. In contrast, lower information cost for day traders, v_d , could reflect regulations that disclose information about market transactions, such as recent changes to stress tests, the use of Central Counterparties (CCPs), disclosures about trading positions, or the use of discount windows. Our analysis shows that regulations that facilitate access to firms’ profitability information make the economy more resilient to recessions and financial shocks. In contrast, those that facilitate information about stock markets’ operations do the opposite.

While [Table VII](#), Panel A shows how alternative economies fare when facing both productivity and liquidity shocks on impact, Panel B shows their differences along their stochastic steady state. For instance, economies with lower information costs exhibit greater overall information acquisition by all traders. However, if information about fundamentals is cheaper, the economy displays higher capital, consumption, and output in a stochastic steady state, whereas the opposite occurs if information about assets’ liquidity is cheaper.

Our exercise naturally abstracts from many channels through which transparency regulations affect markets, hence, it should not be taken as direct policy advice. Yet it helps bring forward potential spillovers from such policies and quantify them.

TABLE VIII
OUTPUT UNDER ALTERNATIVE INFORMATION SOURCES

	Baseline	$\theta_n^s, \varepsilon_n^s$	p^s, ε_n^s	p^s, θ_n^s	ε_n^s	θ_n^s	p^s
Low Z, low γ	0.104	0.089	0.100	0.085	0.077	0.065	0.080
Low Z, high γ	0.101	0.089	0.096	0.080	0.077	0.065	0.074
High Z, low γ	0.119	0.105	0.115	0.097	0.093	0.078	0.091
High Z, high γ	0.117	0.105	0.111	0.094	0.093	0.078	0.087
Stochastic average	0.110	0.097	0.106	0.089	0.085	0.072	0.083

Notes. The first four rows refer to the various aggregate states of the economy, and the last row computes the long-run average output with the stationary distribution. All output figures are computed at the same capital level.

V.C. The Relative Importance of Different Information Sources

The structural nature of our analysis allows us to decompose the importance of learning from the three main information sources in the model: stock prices, private signals on the learnable component of productivity θ_n^s , and private signals about the non-learnable (to traders) component of productivity ε_n^s . This also allows us to interpret our estimates for $\sigma_{\theta_{in}^s}$ and $\sigma_{\varepsilon_{in}^s}$. Table VIII presents the output levels across various states of the economy for alternative combinations of these three sources. Importantly, we keep the aggregate capital stock constant across scenarios to isolate losses due to static misallocation.

Two messages stand out. First, relative to the 0.11 benchmark (Table VIII, first column), the largest output drop is 19% without ε_n^s (to 0.089), followed by 12% without stock prices (to 0.097) and 4% without θ_n^s (to 0.106). Note that ε_n^s is the only signal about one of the two components of productivity, hence losing it is quite costly in terms of allocations. Second, both p^s and θ_n^s are important on their own, with prices conveying a bit more information. Observing prices on top of ε_n^s leads to an increase of output of 25% (from 0.085 to 0.106), while observing private signals about θ_n on top of ε_n^s leads to an increase of output of 14% (from 0.085 to 0.097). In other words, stock prices exhibit an economically significant “revelatory price efficiency,” above and beyond other sources of information.

VI. CONCLUSIONS

Stock markets contribute to the economy in two distinct ways. One is conveying, through prices, useful information about the

best use of resources. The other is allowing agents to access liquidity quickly by trading stocks. We explored how changes in the relevance of the latter affect the performance of the former. When banks (or bond markets) fall short of providing credit, agents rely more on stock markets to access liquidity, eroding the role of stocks in revealing information. To connect the two roles of stock markets, we have provided a model of price formation with endogenous information acquisition and integrated it with a real business cycle model with heterogeneous firms.

In addition, we provide a novel measure of stock price informativeness that captures the extent to which prices reveal information about firms held by market participants. We implement our measure across many countries and periods, enabling us to document their fluctuations and relate them to recessions and banking crises. Because our measure is structural, we can also decompose the sources behind those fluctuations. We show that the informative role of stock markets weakens particularly during periods of distress in funding liquidity. Furthermore, this decline seems mostly related to heightened trading activity unrelated to firms' fundamentals, not to less information acquisition or worse information quality.

These findings highlight a novel link between the functioning of credit markets and stock markets. Even though these markets provide similar services, such as liquidity and information, some argue that banks are more efficient at supplying liquidity and at revealing information about firms (see the discussion in [Gorton and Ordoñez 2023](#)). We argue here that when banks are in distress, stock markets step in to supply liquidity at the expense of their role in revealing information and guiding resource allocation. We assess this link quantitatively by calibrating the full model to the United States. We show that stock markets mitigate the output losses from recessions by 4.4%, but if those recessions are accompanied by financial distress, stock markets go from "friend to foe," magnifying output losses by 22%.

Our approach entails certain limitations that we hope will spark further research. First, our modeling framework assumes the absence of feedback effects from the real economy to financial markets (prices affect investment choices, but investment choices do not affect prices). Adding this two-way feedback would imply losing the tractability provided by linear pricing but would give rise to potential magnification forces. Second, our counterfactual

analysis only centers on how the information environment affects stock price informativeness, abstracting from other potential channels through which information affects production, as in [Plantin and Tirole \(2018\)](#). A more comprehensive study that assesses the overall costs and benefits of information provision would provide a more robust evaluation of the desirability and regulation of information.

SUPPLEMENTARY MATERIAL

An Online Appendix for this article can be found at [The Quarterly Journal of Economics](#) online.

DATA AVAILABILITY

The data and code underlying this article is available in [Bostanci and Ordoñez \(2026\)](#), in the Harvard Dataverse, <https://doi.org/10.7910/DVN/KRMMS7>.

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